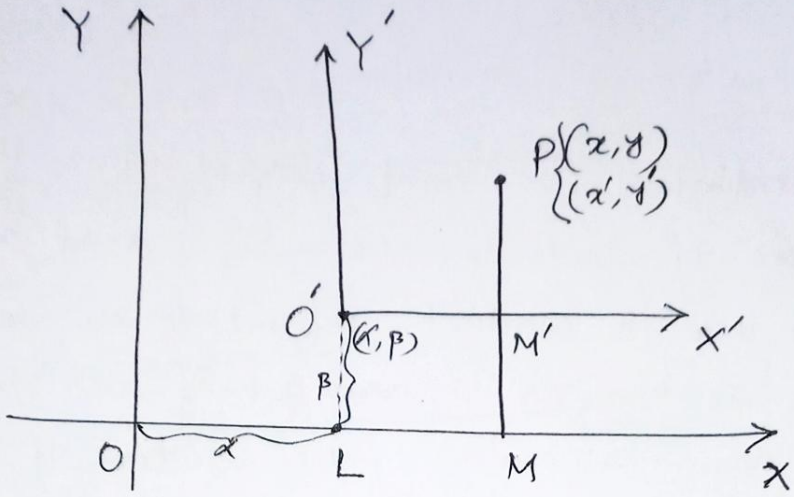
(Transformation of coordinates with respect to change of origin or axes:)

Transformation of coordinates :

The coordinates of a point depend upon the origin and the axes of coordinates chosen. If a point be given in position, then its coordinates referred to one set of axes will change as soon as a new set of axes be taken. That is, the above coordinates of the equation of the curve changes when the origin is changed or the direction of axes changed or both. It is evident that the equation of a curve will also change by such a transformation. But if we know the equation of the curve referred to one set of axes, we can deduce the equation referred to another set of axes. When we pass from one set of axes to another, the process is known as the transformation of coordinates. In other words, the process of changing the coordinates of a point or the equation of a curve is called transformation of coordinates.

To change the origin of coordinates without changing the direction of axes (Translation of axes):

Let, OX and OY be the original axes of coordinates. Let us take a new pair of axes O'X' and O'Y' parallel to the old pair OX and OY; Let, (α, β) be the coordinates of the new origin O' referred to the original axes.



Suppose, P be any point whose coordinates referred to the old axes are $P(x,y)$ and referred to the new axes $P(x',y')$. Draw PM perpendicular to OX, meeting $O'X'$ at M' and OX at M, respectively. From new origin O', draw $O'L$ perpendicular to OX. Then, we have,

$$OL = \alpha, \quad O'L = \beta, \quad OM = x, \quad PM = y$$

Also, $O'M' = x'$ & $PM' = y'$

From the geometry of the figure, we get,

$$OM = OL + LM = OL + O'M'$$

$$\Rightarrow x = \alpha + x'$$

Similarly,

$$PM = M'M + PM' = O'L + PM'$$

$$\Rightarrow y = \beta + y'$$

Therefore, the transformed coordinates are

$$\left.\begin{aligned} x &= x' + \alpha \\ y &= y' + \beta \end{aligned}\right\} \quad \text{——— (1) [old to new]}$$

$$\& \quad \left.\begin{aligned} x' &= x - \alpha \\ y' &= y - \beta \end{aligned}\right\} \quad \text{——— (2) [new to old]}$$

Here, (1) and (2) are the equations of transformation from the old to the new axes and from the new to the old, respectively.

Thus, when the origin is transferred to (α, β), the transformed equation of a curve is obtained by substituting $x'+\alpha$ for x and $y'+\beta$ for y and finally suppressing the accents. That is, the result is obtained only by putting $x+\alpha$ for x and $y+\beta$ for y in the original equation.

—o—

Example:

What does the equation $x^2+y^2-4x-6y+6=0$ become when the origin is transferred to $(2,3)$, the direction of the axes remaining unaltered?

Solution:

The given equation is

$$x^2 + y^2 - 4x - 6y + 6 = 0 \quad \text{——— (1)}$$

Putting $x = x' + 2$ and $y = y' + 3$ into equⁿ ① we have

$$(x'+2)^2 + (y'+3)^2 - 4(x'+2) - 6(y'+3) + 6 = 0$$

$$\Rightarrow x'^2 + 4x' + 4 + y'^2 + 6y' + 9 - 4x' - 8 - 6y' - 18 + 6 = 0$$

$$\Rightarrow x'^2 + y'^2 - 7 = 0$$

Removing suffixes we get

$$x^2 + y^2 - 7 = 0$$

Ans

Example:

Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin is transferred to the point $(2, -1)$.

Solution:

The given equation is

$$2x^2 + 3y^2 - 8x + 6y - 7 = 0 \quad \text{——————} \quad ①$$

Putting $x = x' + 2$ and $y = y' - 1$ into equⁿ ① we obtain

$$2(x'+2)^2 + 3(y'-1)^2 - 8(x'+2) + 6(y'-1) - 7 = 0$$

$$\Rightarrow 2(x'^2 + 4x' + 4) + 3(y'^2 - 2y' + 1) - 8x' - 16 + 6y' - 6 - 7 = 0$$

$$\Rightarrow 2x'^2 + 8x' + 8 + 3y'^2 - 6y' + 3 - 8x' + 6y' - 29 = 0$$

$$\Rightarrow 2x'^2 + 3y'^2 = 18$$

Removing suffixes we have

$$2x^2 + 3y^2 = 18$$

(Ans)

Example:

Determine the equation of the curve $2x^2 + y^2 - 4x + 4y = 0$ when the origin is transferred to the point $(1, -2)$.

Solution:

The given equation is

$$2x^2 + y^2 - 4x + 4y = 0 \quad \text{——————} \quad (1)$$

Putting $x = x' + 1$ & $y = y' - 2$ into eqn (1) we obtain

$$2(x'+1)^2 + (y'-2)^2 - 4(x'+1) + 4(y'-2) = 0$$

$$\Rightarrow 2(x'^2 + 2x' + 1) + y'^2 - 4y' + 4 - 4x' - 4 + 4y' - 8 = 0$$

$$\Rightarrow 2x'^2 + 4x' + 2 + y'^2 - 4y' - 4x' + 4y' - 8 = 0$$

$$\Rightarrow 2x'^2 + y'^2 - 6 = 0$$

$$\Rightarrow 2x'^2 + y'^2 = 6$$

Removing suffixes we can write

$$2x^2 + y^2 = 6$$

(Ans)

Example:

Determine the equation of the curve $x^2-6x+2y^2+7=0$ when the origin is transferred to the point $(3,1)$.

Solution:

The given equation is

$$x^2-6x+2y^2+7=0 \quad \text{——————} (1)$$

Putting $x=x'+3$ & $y=y'+1$ into equ^n (1) we obt

$$(x'+3)^2-6(x'+3)+2(y'+1)^2+7=0$$

$$\Rightarrow x'^2+6x'+9-6x'-18+2y'^2+4y'+2+7=0$$

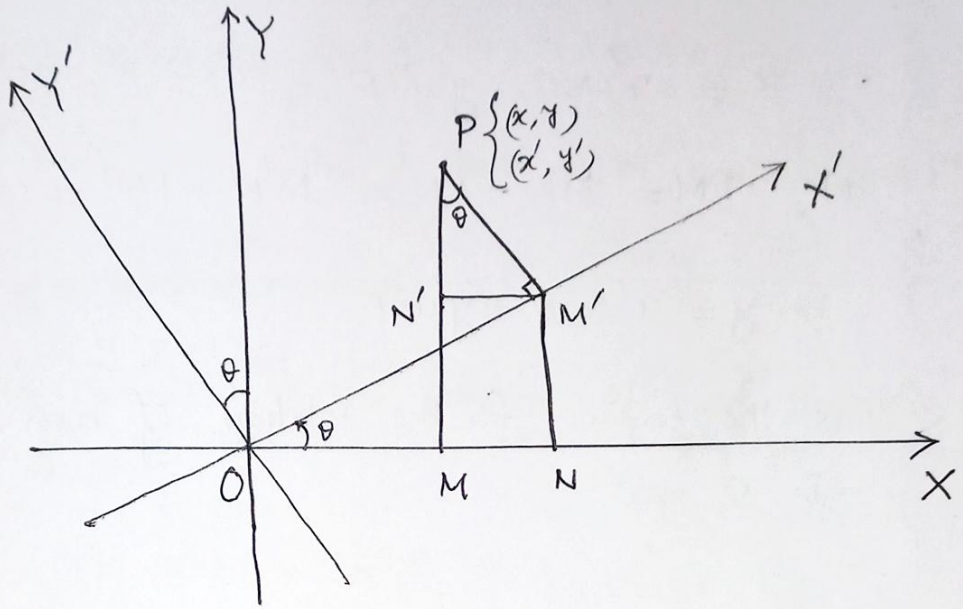
$$\Rightarrow x'^2+2y'^2+4y'=0$$

Removing suffixes we can write

$$x^2+2y^2+4y=0$$

Ans

To transform from one set of rectangular axes system to another with the same origin (Rotation of axes):



Let, OX and OY be the original axes of coordinates, and OX' and OY' be the new axes through the same origin and θ be the angle through which they have been turned in the same sense. That is $\angle XOX' = \angle YOY' = \theta$.

Let P be any point whose coordinates are (x, y) referred to the original axes and (x', y') referred to the new axes of coordinates. Draw PM and PM' perpendiculars to OX and OX', respectively. Draw M'N parallel to OY cutting OX at N and M'N' parallel to OX cutting PM at N'. Then we can write,

$OM = x$; $OM' = x'$; $PM = y$; $PM' = y'$;

Again, $PM \perp OX$ and $PM' \perp OX'$

$\therefore \angle MPM' = \angle XOX' = \theta$

Now, using the geometry of the figure, we have

$$OM = ON - MN = OM'\cos\theta - PM'\sin\theta$$

$$\Rightarrow x = x'\cos\theta - y'\sin\theta$$

And, $PM = MN' + PN' = M'N + PN' = OM'\sin\theta + PM'\cos\theta$

$$\Rightarrow y = x'\sin\theta + y'\cos\theta$$

Thus, the formula for the rotation of axes through an angle θ are:

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (1)}$$

Solving the equations in (1), we get

$$\left.\begin{aligned} x' &= x\cos\theta + y\sin\theta \\ y' &= -x\sin\theta + y\cos\theta \end{aligned}\right\} \quad \text{——— (2)}$$

These transforming equations (1) & (2) may be conveniently remembered from the following scheme:

	x'	y'
x	$\cos\theta$	$-\sin\theta$
y	$\sin\theta$	$\cos\theta$

which may be read either horizontally or vertically. The transformed equation of a curve, in this transformation, is then obtained by substituting $x\cos\theta - y\sin\theta$ for x and $x\sin\theta + y\cos\theta$ for y.

Change of origin with the change of the direction of axes:

If the origin is first transferred to the point (α, β) and the new set of axes through this origin be inclined at an angle θ to the original axes, the formulae of transformation will obviously be the combination of the above two sets of formulae, i.e.,

$$\left.\begin{aligned} x &= \alpha + x'\cos\theta - y'\sin\theta \\ \text{and } y &= \beta + x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (3)}$$

Example:

Transform to axes inclined at 45° to the original axes of the following equations:

(i) $x^2 - y^2 = a^2$

(ii) $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$

Solution:

(i) Given that,

$$x^2 - y^2 = a^2 \quad \text{——— (1)}$$

When the axes have been rotated through an angle 45°, then we have,

$$x = x'\cos 45^\circ - y'\sin 45^\circ = \frac{1}{\sqrt{2}}(x' - y')$$

$$y = x'\sin 45^\circ + y'\cos 45^\circ = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x & y into eqn (1), we get

$$\left\{\frac{1}{\sqrt{2}}(x' - y')\right\}^2 - \left\{\frac{1}{\sqrt{2}}(x' + y')\right\}^2 = a^2$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 + 2x'y' + y'^2) = a^2$$

$$\Rightarrow \cancel{\frac{1}{2}x'^2} - x'y' + \cancel{\frac{1}{2}y'^2} - \cancel{\frac{1}{2}x'^2} - x'y' - \cancel{\frac{1}{2}y'^2} = a^2$$

$$\Rightarrow -2x'y' = a^2$$

$$\Rightarrow 2x'y' + a^2 = 0$$

Now dropping the suffixes, the equation becomes

$$2xy + a^2 = 0 \quad \text{——— (2)}$$

Hence, equⁿ (2) is the required transformed equation.

Ans

(ii) Given that,

$$x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0 \quad \text{——— (1)}$$

When the axes have been rotated through an angle 45°, then we have,

$$x = x'\cos 45^\circ - y'\sin 45^\circ = \frac{1}{\sqrt{2}}(x' - y')$$

$$y = x'\sin 45^\circ + y'\cos 45^\circ = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x and y into equⁿ (1), we get

$$\left\{\frac{1}{\sqrt{2}}(x'-y')\right\}^2 - \left\{\frac{1}{\sqrt{2}}(x'+y')\right\}^2 - 2\sqrt{2}\cdot\frac{1}{\sqrt{2}}(x'-y') - 10\sqrt{2}\cdot\frac{1}{\sqrt{2}}(x'+y') + 2 = 0$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 + 2x'y' + y'^2) - 2x' + 2y' - 10x' - 10y' + 2 = 0$$

$$\Rightarrow \cancel{\frac{1}{2}x'^2} - x'y' + \cancel{\frac{1}{2}y'^2} - \cancel{\frac{1}{2}x'^2} - x'y' - \cancel{\frac{1}{2}y'^2} - 12x' - 8y' + 2 = 0$$

$$\Rightarrow -2x'y' - 12x' - 8y' + 2 = 0$$

$$\Rightarrow x'y' + 6x' + 4y' = 1$$

Now dropping the suffixes, the equation becomes,

$$xy + 6x + 4y = 1 \quad \text{——— (2)}$$

Thus, equⁿ (2) is the required transformed equation.

(Ans)

Example:

Suppose that the axes of an xy-coordinate system are rotated through an angle of $\theta = 45°$ to obtain an $x'y'$-coordinate system. Find the equation of the curve $x^2 - xy + y^2 - 6 = 0$ in $x'y'$-coordinates.

Solution:

Given that,

$$x^2 - xy + y^2 - 6 = 0 \quad \text{——— (1)}$$

If an xy-coordinate system is gone through a rotation of $45°$ of the axes, then we have

$$x = x'\cos 45° - y'\sin 45° = \frac{1}{\sqrt{2}}(x' - y')$$

& $$y = x'\sin 45° + y'\cos 45° = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x & y into the given equⁿ (1) yields

$$\left\{\frac{1}{\sqrt{2}}(x' - y')\right\}^2 - \frac{1}{\sqrt{2}}(x' - y') \cdot \frac{1}{\sqrt{2}}(x' + y') + \left\{\frac{1}{\sqrt{2}}(x' + y')\right\}^2 - 6 = 0$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 - y'^2) + \frac{1}{2}(x'^2 + 2x'y' + y'^2) - 6 = 0$$

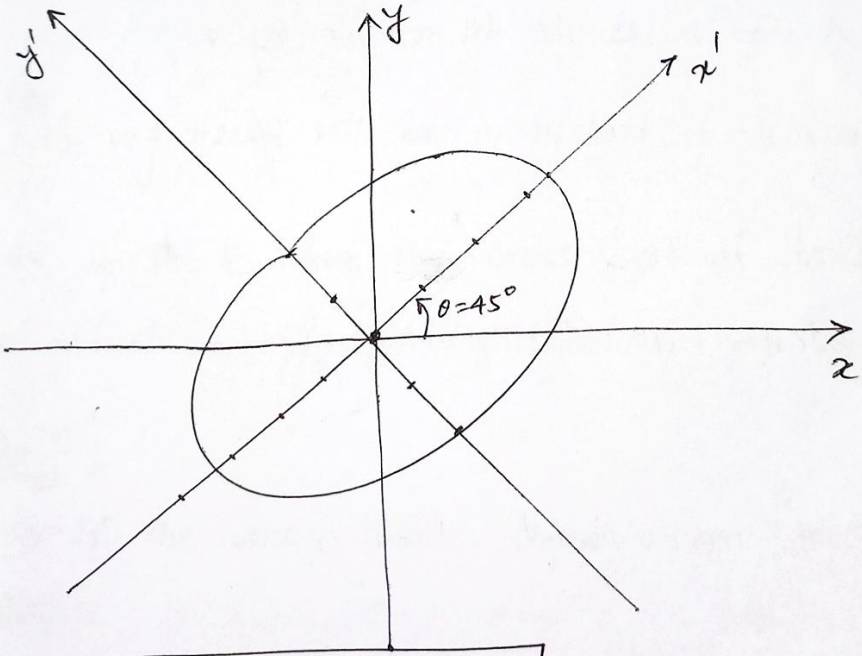
$$\Rightarrow \frac{1}{2}\left(x'^2 - 2x'y' + y'^2 - x'^2 + y'^2 + x'^2 + 2x'y' + y'^2\right) - 6 = 0$$

$$\Rightarrow \frac{1}{2}\left(x'^2 + 3y'^2\right) - 6 = 0$$

$$\Rightarrow x'^2 + 3y'^2 = 12$$

$$\Rightarrow \frac{x'^2}{12} + \frac{y'^2}{4} = 1$$

which is the equation of an ellipse shown in the following figure.



$$\boxed{x^2 - xy + y^2 - 6 = 0}$$

Theorem:

If the equation $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ ——— (1) is such that $B\neq 0$, and if an $x'y'$-coordinate system is obtained by rotating the xy-axes through an angle θ, then prove that,

(a) the expression (1) changes to

$$A'x'^2+B'x'y'+C'y'^2+D'x'+E'y'+F'=0$$

(b) $B^2-4AC=B'^2-4A'C'$ for all values of θ.

(c) $A+C=A'+C'$ for all values of θ.

(d) xy term will be vanished if $\cot 2\theta=\frac{A-C}{B}$

(e) if $A=C$ then the cross product term can be eliminated by rotating through an angle $\theta=45^\circ$.

Proof:

(a) If the axes rotate through an angle θ, then we have

$$x=x'\cos\theta-y'\sin\theta$$

& $$y=x'\sin\theta+y'\cos\theta$$

Substituting the values of x & y into equ^n (1) yields

$$A(x'\cos\theta-y'\sin\theta)^2+B(x'\cos\theta-y'\sin\theta)(x'\sin\theta+y'\cos\theta)$$
$$+C(x'\sin\theta+y'\cos\theta)^2+D(x'\cos\theta-y'\sin\theta)+E(x'\sin\theta+y'\cos\theta)+F$$

$$\Rightarrow A(x'^2\cos^2\theta - x'y'\sin 2\theta + y'^2\sin^2\theta)$$
$$+ B(x'^2\sin\theta\cos\theta - x'y'\sin^2\theta + x'y'\cos^2\theta - y'^2\sin\theta\cos\theta)$$
$$+ C(x'^2\sin^2\theta + x'y'\sin 2\theta + y'^2\cos^2\theta)$$
$$+ D(x'\cos\theta - y'\sin\theta) + E(x'\sin\theta + y'\cos\theta) + F = 0$$

$$\Rightarrow (A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta)x'^2$$
$$+ \{B(\cos^2\theta - \sin^2\theta) + (C - A)\sin 2\theta\}x'y'$$
$$+ (A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)y'^2$$
$$+ (D\cos\theta + E\sin\theta)x' + (-D\sin\theta + E\cos\theta)y' + F = 0$$

$$\Rightarrow A'x'^2 + B'x'y' + C'y'^2 + D'x' + E'y' + F' = 0$$

where,

$$A' = A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta$$

$$B' = B\cos 2\theta + (C - A)\sin 2\theta$$

$$C' = A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta$$

$$D' = D\cos\theta + E\sin\theta$$

$$E' = -D\sin\theta + E\cos\theta$$

$$F' = F$$

(b) $B'^2 - 4A'C' = \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- 4(A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta)$.

$(A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- 2(A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta)$

$\cdot 2(A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- (A\cdot 2\cos^2\theta + B\cdot 2\sin\theta\cos\theta + C\cdot 2\sin^2\theta)$

$\cdot (A\,2\sin^2\theta - B\cdot 2\sin\theta\cos\theta + C\cdot 2\cos^2\theta)$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- (A(1+\cos 2\theta) + B\sin 2\theta + C(1-\cos 2\theta))$

$\cdot (A(1-\cos 2\theta) - B\sin 2\theta + C(1+\cos 2\theta))$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- \{(A+C) + (A-C)\cos 2\theta + B\sin 2\theta\}\{(A+C) - (A-C)\cos 2\theta - B\sin 2\theta\}$

$$= \left\{B\cos 2\theta + (C-A)\sin 2\theta\right\}^2$$

$$-\left[(A+C) + \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}\right]$$

$$\cdot\left[(A+C) - \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}\right]$$

$$= B^2\cos^2 2\theta + 2B(C-A)\sin 2\theta\cdot\cos 2\theta + (C-A)^2\sin^2 2\theta$$

$$-\left[(A+C)^2 - \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}^2\right]$$

$$= B^2\cos^2 2\theta + 2B(C-A)\sin 2\theta\cdot\cos 2\theta + (C-A)^2\sin^2 2\theta$$

$$-(A+C)^2 + \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}^2$$

$$= B^2\cos^2 2\theta + 2B(C-A)\sin 2\theta\cos 2\theta + (C-A)^2\sin^2 2\theta$$

$$-(A+C)^2 + (A-C)^2\cos^2 2\theta + 2B(A-C)\sin 2\theta\cos 2\theta$$

$$+ B^2\sin^2 2\theta$$

$$= B^2(\cos^2 2\theta + \sin^2 2\theta) - \cancel{2B(A-C)\sin 2\theta\cdot\cos 2\theta}$$

$$+(C-A)^2(\sin^2 2\theta + \cos^2 2\theta) - (A+C)^2$$

$$\cancel{+\,2B(A-C)\sin 2\theta\cos 2\theta}$$

$$= B^2 + (C-A)^2 - (C+A)^2$$

$$= B^2 - \left[(C+A)^2 - (C-A)^2\right]$$

$\therefore B'^2 - 4A'C' = B^2 - 4AC$

(Proved)

(c) $A' + C' = A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta$

$+ A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta$

$\Rightarrow A' + C' = A(\cos^2\theta + \sin^2\theta) + C(\sin^2\theta + \cos^2\theta)$

$\therefore A' + C' = A + C$

(Proved)

(d) xy term will be vanished if

$B' = 0$

$\Rightarrow B\cos 2\theta + (C - A)\sin 2\theta = 0$

$\Rightarrow B\cos 2\theta = (A - C)\sin 2\theta$

$\Rightarrow \cot 2\theta = \frac{A - C}{B}$

$\therefore \cot 2\theta = \frac{A - C}{B}$

(Showed)

(e) We have already proved that

$\cot 2\theta = \frac{A - C}{B}$ ——— (ii)

If $A = C$, then equⁿ (11) becomes

$$\cot 2\theta = \frac{0}{B} = 0 = \cot 90^\circ$$

$$\Rightarrow \quad 2\theta = 90^\circ$$

$$\therefore \quad \theta = 45^\circ$$

(Proved)

Example:

Identify and sketch the curve

$$153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0$$

Solution:

Given that,

$$153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0 \quad \text{——— (1)}$$

Here we have,

$$A = 153; \quad B = -192; \quad C = 97$$

$$\therefore \cot 2\theta = \frac{A - C}{B} = \frac{153 - 97}{-192} = -\frac{7}{24}$$

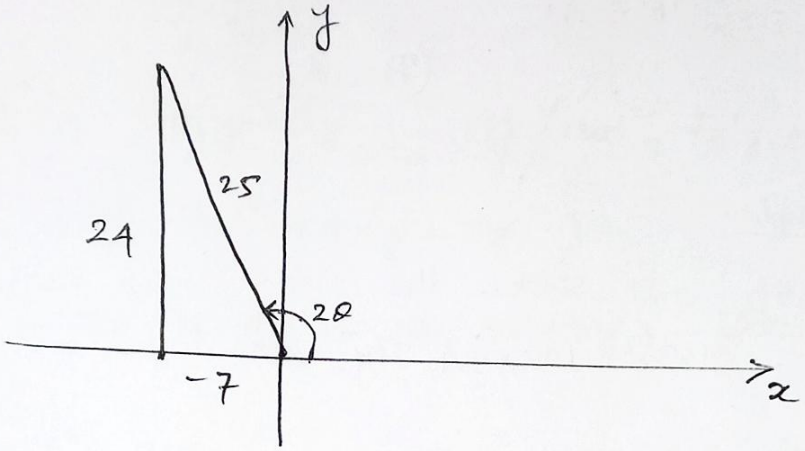
$$\Rightarrow \cot 2\theta = -7/24$$

Since θ is to be chosen in the range $0 < \theta < \pi/2$, this relationship is represented by the triangle shown in fig. 1. From that triangle we obtain $\cos 2\theta = -7/25$, which

implies that

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = 3/5$$

$$\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1+7/25}{2}} = 4/5$$



We know that

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \longrightarrow (2)$$

Substituting the values of $\cos\theta$ & $\sin\theta$ into eqn (2), we have,

$$\left.\begin{aligned} x &= \frac{3}{5}x' - \frac{4}{5}y' \\ \& \quad y &= \frac{4}{5}x' + \frac{3}{5}y' \end{aligned}\right\} \longrightarrow (3)$$

Substituting the values of (3) into (1) yields

$$\frac{153}{25}(3x'-4y')^2 - \frac{192}{25}(3x'-4y')(4x'+3y') + \frac{97}{25}(4x'+3y')^2 - \frac{30}{5}(3x'-4y') - \frac{40}{5}(4x'+3y') - 200 = 0$$

$$\Rightarrow \frac{153}{25}(9x'^2 - 24x'y' + 16y'^2) - \frac{192}{25}(12x'^2 - 16x'y' + 9x'y' - 12y'^2)$$
$$+ \frac{97}{25}(16x'^2 + 24x'y' + 9y'^2)$$
$$- \frac{30}{5}(3x' - 4y') - \frac{40}{5}(4x' + 3y') - 200 = 0$$

$$\Rightarrow 153(9x'^2 - 24x'y' + 16y'^2) - 192(12x'^2 - 7x'y' - 12y'^2)$$
$$+ 97(16x'^2 + 24x'y' + 9y'^2)$$
$$- 150(3x' - 4y') - 200(4x' + 3y') - 5000 = 0$$

$$\Rightarrow 1377x'^2 - 3672x'y' + 2448y'^2 - 2304x'^2 + 1344x'y'$$
$$+ 2304y'^2 + 1552x'^2 + 2328x'y' + 873y'^2$$
$$- 450x' + 600y' - 800x' - 600y' - 5000 = 0$$

$$\Rightarrow 625x'^2 + 5625y'^2 - 1250x' - 5000 = 0$$

$$\Rightarrow 25x'^2 + 225y'^2 - 50x' - 200 = 0$$

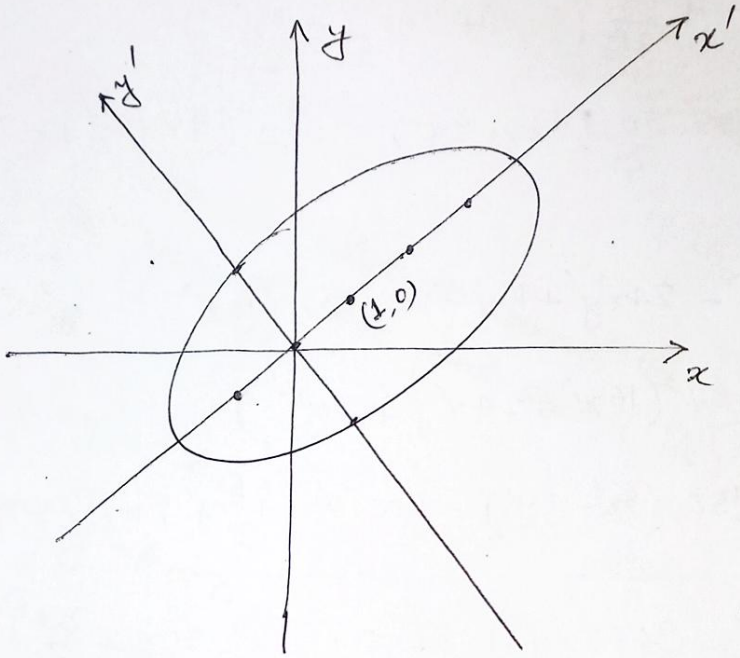
$$\Rightarrow x'^2 + 9y'^2 - 2x' - 8 = 0$$

$$\Rightarrow x'^2 - 2x' + 1 + 9y'^2 - 9 = 0$$

$$\Rightarrow (x' - 1)^2 + 9y'^2 = 9$$

$$\Rightarrow \frac{(x' - 1)^2}{9} + y'^2 = 1$$

$$\Rightarrow \frac{(x'-1)^2}{3^2} + \frac{y'^2}{1^2} = 1 \qquad \text{——— (4)}$$



Thus, equⁿ (4) is the equation in the $x'y'$– coordinate syste of an ellipse with centre $(1,0)$ in that coordinate system and the length of the semi major axis $a = 3$ and that of the semi minor axis $b = 1$.

Th^m :

Transform the general eqⁿ of second degree $ax^2+2hxy+by^2+2gx+2fy+c=0$ to one in which there is no term involving x, y, and xy, both set of axes being rectangular.

Proof :

Given, $ax^2+2hxy+by^2+2gx+2fy+c=0$ ——— (1)

If the origin is shifted to the point (α, β) without changing the direction of the axes, then we have

$$a(x+\alpha)^2+2h(x+\alpha)(y+\beta)+b(y+\beta)^2+2g(x+\alpha)$$
$$+2f(y+\beta)+c=0$$

$$\Rightarrow ax^2+2hxy+by^2+2x(a\alpha+h\beta+g)+2y(h\alpha+b\beta+f)$$
$$+a\alpha^2+2h\alpha\beta+b\beta^2+2g\alpha+2f\beta+c=0 \quad \text{——— (2)}$$

Now, x & y terms will be removed if we set,

$$\left.\begin{aligned} a\alpha+h\beta+g&=0 \\ \& \; h\alpha+b\beta+f&=0 \end{aligned}\right\} \quad \text{——— (3)}$$

Solving (3) for α & β we obtain

$$\alpha = \frac{hf - bg}{ab - h^2}; \quad \beta = \frac{gh - af}{ab - h^2}$$

Now, eqⁿ (2) becomes

$$ax^2 + 2hxy + by^2 + q = 0 \qquad \text{——— (3)}$$

where,

$$q = a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c$$

$$= \alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + c$$

$$= \alpha \cdot 0 + \beta \cdot 0 + g\alpha + f\beta + c$$

$$\Rightarrow q = g\alpha + f\beta + c$$

Hence, the general eqⁿ of second degree having no terms involving x & y becomes

$$ax^2 + 2hxy + by^2 + q_1 = 0$$

where, $q = g\alpha + f\beta + c$ or, $q = \alpha g + \beta f + c$

Now if the set of axes is rotated through an angle θ then eqⁿ (3) becomes

$$a(x\cos\theta - y\sin\theta)^2 + 2h(x\cos\theta - y\sin\theta)(x\sin\theta + y\cos\theta) + b(x\sin\theta + y\cos\theta)^2 + q = 0$$

CamScanner

$$\Rightarrow x^2(a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta)$$
$$+2xy(-a\sin\theta\cos\theta + h\cos^2\theta - h\sin^2\theta + b\sin\theta\cos\theta)$$
$$+y^2(a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta) + c_1 = 0$$

$$\Rightarrow x^2(a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta)$$
$$+2xy\{h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta\}$$
$$+y^2(a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta) + c_1 = 0$$

$$\Rightarrow x^2(a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta)$$
$$+2xy\left\{h\cos 2\theta - \left(\frac{a-b}{2}\right)\sin 2\theta\right\}$$
$$+y^2(a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta) + c_1 = 0$$
$$\text{---------- (4)}$$

Now the xy-term will be removed if we set

$$h\cos 2\theta - \left(\frac{a-b}{2}\right)\sin 2\theta = 0$$

$$\Rightarrow \left(\frac{a-b}{2}\right)\sin 2\theta = h\cos 2\theta$$

$$\Rightarrow \tan 2\theta = \frac{2h}{a-b} \Rightarrow \tan 2\theta = \tan^{-1}\left(\frac{2h}{a-b}\right)$$

$$\Rightarrow \theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$$

Let us consider equ^{n} (4) to be

$$a_1x^2 + b_1y^2 + c_1 = 0$$

where,

$$a_1 = a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta.$$

$$b_1 = a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta$$

& $$\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$$

Now,

$$a_1 + b_1 = a(\cos^2\theta + \sin^2\theta) + b(\sin^2\theta + \cos^2\theta)$$

$$= a + b$$

Again,

$$2a_1 = a(2\cos^2\theta) + 2h.(2\sin\theta\cos\theta) + b\,2\sin^2\theta$$

$$= a(1+\cos2\theta) + 2h\sin2\theta + b(1-\cos2\theta)$$

$$\Rightarrow 2a_1 = (a+b) + \{(a-b)\cos2\theta + 2h\sin2\theta\} \quad \text{---(5)}$$

Also,

$$2b_1 = a(2\sin^2\theta) - 2h(2\sin\theta\cos\theta) + b(2\cos^2\theta)$$

$$= a(1-\cos2\theta) - 2h\sin2\theta + b(1+\cos2\theta)$$

CamScanner

$$\Rightarrow 2b_1 = (a+b) - \{(a-b)\cos 2\theta + 2h\sin 2\theta\} \quad \text{---(6)}$$

(5) × (6)

$$4a_1b_1 = (a+b)^2 - \{(a-b)\cos 2\theta + 2h\sin 2\theta\}^2$$

$$= (a+b)^2 - [(a-b)^2\cos^2 2\theta + 2(a-b).2h\cos 2\theta\sin 2\theta + 4h^2\sin^2 2\theta]$$

$$= (a+b)^2 - (a-b)^2\cos^2 2\theta - 4(a-b)h\cos 2\theta\sin 2\theta - 4h^2\sin^2 2\theta$$

$$= (a+b)^2 - (a-b)^2(1-\sin^2 2\theta) - 4(a-b)h\cos 2\theta\sin 2\theta - 4h^2\sin^2 2\theta$$

$$= (a+b)^2 - (a-b)^2 + (a-b)^2\sin^2 2\theta - 4(a-b)h\cos 2\theta\sin 2\theta - 4h^2(1-\cos^2 2\theta)$$

$$= 4ab + (a-b)^2\sin^2 2\theta - 4(a-b)h\cos 2\theta\sin 2\theta - 4h^2 + 4h^2\cos^2 2\theta$$

$$= 4ab - 4h^2 + [(a-b)^2\sin^2 2\theta - 2.(a-b)2h\sin 2\theta\cos 2\theta + (2h)^2\cos^2 2\theta]$$

$$\Rightarrow 4a_1b_1 = 4(ab-h^2) + \left[\{(a-b)\sin 2\theta\}^2 + 2(a-b)\sin 2\theta \cdot 2h\cos 2\theta + (2h\cos 2\theta)^2\right]$$

$$\Rightarrow 4a_1b_1 = 4(ab-h^2) + \left[(a-b)\sin 2\theta - 2h\cos 2\theta\right]^2$$

$$\Rightarrow 4a_1b_1 = 4(ab-h^2) + \left[2h\cos 2\theta - (a-b)\sin 2\theta\right]^2$$

$$= 4(ab-h^2) + 0^2 \quad [\because 2h\cos 2\theta - (a-b)\sin 2\theta = 0]$$

$$\Rightarrow \boxed{a_1b_1 = ab-h^2}$$

Hence, the required equⁿ where xy-term is missing can be written as

$$a_1x^2 + b_1y^2 + c = 0$$

where, $a_1 + b_1 = a + b$

& $a_1b_1 = ab - h^2$

Th^m: (Invarient Theorem)

If the transformed equation of $ax^2+2hxy+by^2=0$ is $a_1x^2+2h_1xy+b_1y^2=0$ when the direction of axes is turned through an angle θ without changing the origin, then show that $a+b$ and $ab-h^2$ are invarient; i.e., $a+b=a_1+b_1$ and $ab-h^2=a_1b_1-h_1^2$.

Proof:

Given, $ax^2+2hxy+by^2=0$ ——— (1)

Now, if the set of axes is rotated through an angle θ, eqn (1) becomes

$$a(x\cos\theta-y\sin\theta)^2+2h(x\cos\theta-y\sin\theta)(x\sin\theta+y\cos\theta)+b(x\sin\theta+y\cos\theta)^2=0$$

$$\Rightarrow x^2(a\cos^2\theta+2h\cos\theta\sin\theta+b\sin^2\theta)+2xy(-a\sin\theta\cos\theta+h\cos^2\theta-h\sin^2\theta+b\sin\theta\cos\theta)+y^2(a\sin^2\theta-2h\sin\theta\cos\theta+b\cos^2\theta)=0$$

$$\Rightarrow a_1x^2+2h_1xy+b_1y^2=0 \quad \text{——— (2)}$$

where,

$$a_1 = a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta$$

$$h_1 = -a\sin\theta\cos\theta + h\cos^2\theta - h\sin^2\theta + b\sin\theta\cos\theta$$

$$b_1 = a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta$$

Now,

$$a_1 + b_1 = a(\cos^2\theta + \sin^2\theta) + b(\sin^2\theta + \cos^2\theta)$$

$$\Rightarrow a_1 + b_1 = a + b \quad \text{---(3)}$$

(Proved)

Again,

$$2a_1 = 2a\cos^2\theta + 2h(2\cos\theta\sin\theta) + 2b\sin^2\theta$$

$$= a(1+\cos 2\theta) + 2h\sin 2\theta + b(1-\cos 2\theta)$$

$$\therefore 2a_1 = (a+b) + \{(a-b)\cos 2\theta + 2h\sin 2\theta\} \quad \text{---(3)}$$

Also,

$$2b_1 = 2a\sin^2\theta - 2h(2\sin\theta\cos\theta) + 2b\cos^2\theta$$

$$= a(1-\cos 2\theta) - 2h\sin 2\theta + b(1+\cos 2\theta)$$

$$\therefore 2b_1 = (a+b) - \{(a-b)\cos 2\theta + 2h\sin 2\theta\} \quad \text{---(4)}$$

Now,

$$4a_1b_1 = (a+b)^2 - \{(a-b)\cos 2\theta + 2h\sin 2\theta\}^2$$

$$\Rightarrow 4a_1b_1 = (a+b)^2 - \left[(a-b)^2\cos^2 2\theta + 2(a-b)2h\sin 2\theta\cos 2\theta + 4h^2\sin^2 2\theta\right] \quad \text{——— (5)}$$

Again,

$$2h_1 = -2a\sin\theta\cos\theta + 2h(\cos^2\theta - \sin^2\theta) + 2b\sin\theta\cos\theta$$

$$\Rightarrow 2h_1 = 2h\cos 2\theta - (a-b)\sin 2\theta$$

$$\Rightarrow 4h_1^2 = 4h^2\cos^2 2\theta - 2\cdot 2h\cos 2\theta(a-b)\sin 2\theta + (a-b)^2\sin^2 2\theta$$

$$\Rightarrow 4h_1^2 = 4h^2\cos^2 2\theta - 2(a-b)2h\sin 2\theta\cos 2\theta + (a-b)^2\sin^2 2\theta \quad \text{——— (6)}$$

(5) − (6)

$$4a_1b_1 - 4h_1^2 = (a+b)^2 - (a-b)^2\cos^2 2\theta - \cancel{2(a-b)h\sin 4\theta} - 4h^2\sin^2 2\theta - 4h^2\cos^2 2\theta + \cancel{2(a-b)h\sin 4\theta} - (a-b)^2\sin^2 2\theta$$

$$= (a+b)^2 - (a-b)^2\left[\cos^2 2\theta + \sin^2 2\theta\right] - 4h^2\left[\sin^2 2\theta + \cos^2 2\theta\right]$$

$$\Rightarrow 4(a_1 b_1 - h_1^2) = (a+b)^2 - (a-b)^2 - 4h^2$$

$$\Rightarrow 4(a_1 b_1 - h_1^2) = 4ab - 4h^2 = 4(ab - h^2)$$

$$\Rightarrow a_1 b_1 - h_1^2 = ab - h^2$$

(Proved)

Problem:

Rotate the coordinate axes to remove the xy-term. Then identify the type of conic and sketch its graph.

(i) $xy = -9$

(ii) $x^2 - xy + y^2 - 2 = 0$

(iii) $x^2 + 4xy - 2y^2 - 6 = 0$

(iv) $31x^2 + 10\sqrt{3}xy + 21y^2 - 144 = 0$

(v) $x^2 + 2\sqrt{3}xy + 3y^2 + 2\sqrt{3}x - 2y = 0$

(vi) $34x^2 - 24xy + 41y^2 - 25 = 0$

(vii) $9x^2 - 24xy + 16y^2 - 80x - 60y + 100 = 0$

(viii) $5x^2 - 6xy + 5y^2 - 8\sqrt{2}x + 8\sqrt{2}y = 8$

(ix) $52x^2 - 72xy + 73y^2 + 40x + 30y - 75 = 0$

(x) $6x^2 + 24xy - y^2 - 12x + 26y + 11 = 0$

Solution:

(i) Given that,

$$xy = -9 \quad \text{——} \quad (1)$$

Here we have

$$A = 0$$

$$B = 1$$

$$C = 0$$

$$F = 9$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{0}{1} = 0 = \cot \pi/2$$

$$\Rightarrow 2\theta = \pi/2 \quad \therefore \theta = \pi/4$$

We know that

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{---------- (2)}$$

Substituting the value of θ into eqn (2) we get.

$$x = x'\cos\pi/4 - y'\sin\pi/4 = \frac{1}{\sqrt{2}}(x'-y')$$

$$y = x'\sin\pi/4 + y'\cos\pi/4 = \frac{1}{\sqrt{2}}(x'+y')$$

Substituting the values of x & y into eqn (1) we have

$$\frac{1}{\sqrt{2}}(x'-y')\cdot\frac{1}{\sqrt{2}}(x'+y') = -9$$

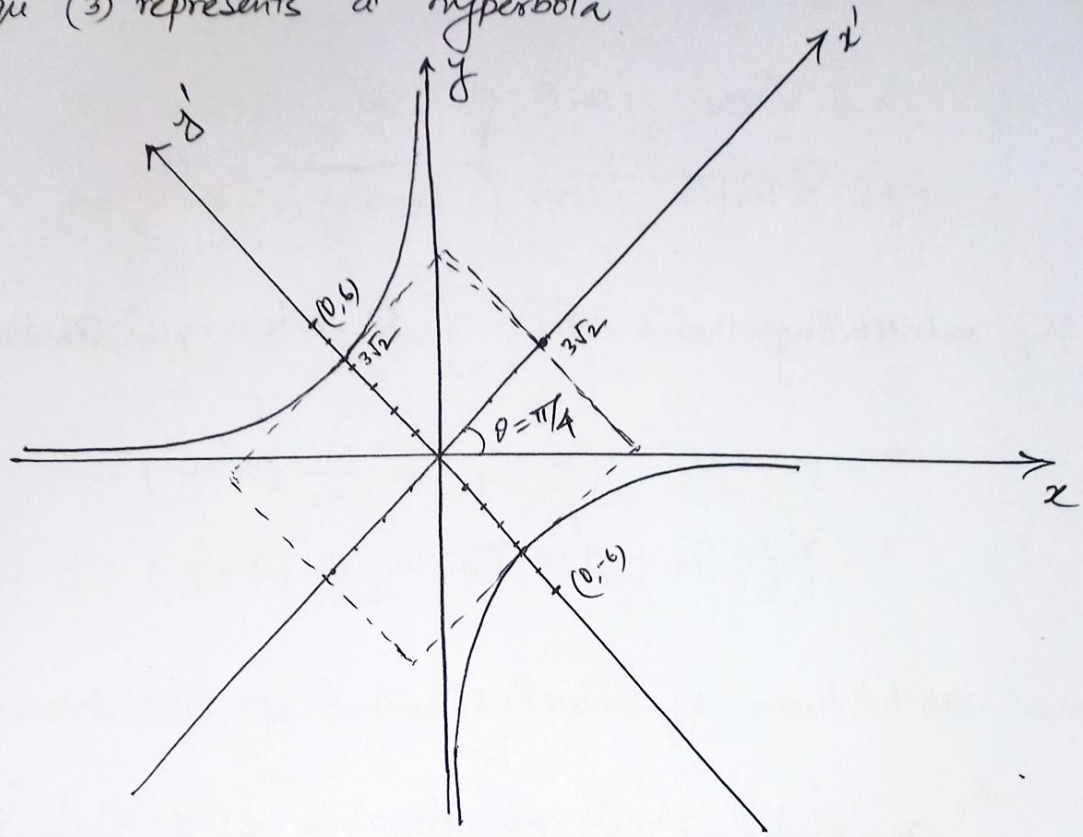
$$\Rightarrow \frac{1}{2}(x'^2 - y'^2) = -9$$

$$\Rightarrow y'^2 - x'^2 = 18$$

$$\Rightarrow \frac{y'^2}{18} - \frac{x'^2}{18} = 1$$

$$\Rightarrow \frac{y'^2}{(3\sqrt{2})^2} - \frac{x'^2}{(3\sqrt{2})^2} = 1 \quad \text{---------- (3)}$$

Equⁿ (3) represents a hyperbola



Solution: (ii)

Given that,

$$x^2 - xy + y^2 - 2 = 0 \quad \longrightarrow (1)$$

Comparing with the standard equation we have

$$A = 1,\ B = -1,\ C = 1,\ F = -2$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{1-1}{-1} = 0 = \cot \pi/2$$

$$\Rightarrow 2\theta = \pi/2$$

$$\therefore \theta = \pi/4 .$$

We know that,

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (2)}$$

Substituting the values of θ into eqn (2) we get

$$x = x'\cos\pi/4 - y'\sin\pi/4 = \frac{1}{\sqrt{2}}(x'-y')$$

$$y = x'\sin\pi/4 + y'\cos\pi/4 = \frac{1}{\sqrt{2}}(x'+y')$$

Substituting the values of x & y into (1) we have

$$\left\{\frac{1}{\sqrt{2}}(x'-y')\right\}^2 - \frac{1}{2}(x'-y')(x'+y') + \left\{\frac{1}{\sqrt{2}}(x'+y')\right\}^2 - 2 = 0$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 - y'^2) + \frac{1}{2}(x'^2 + 2x'y' + y'^2) - 2 = 0$$

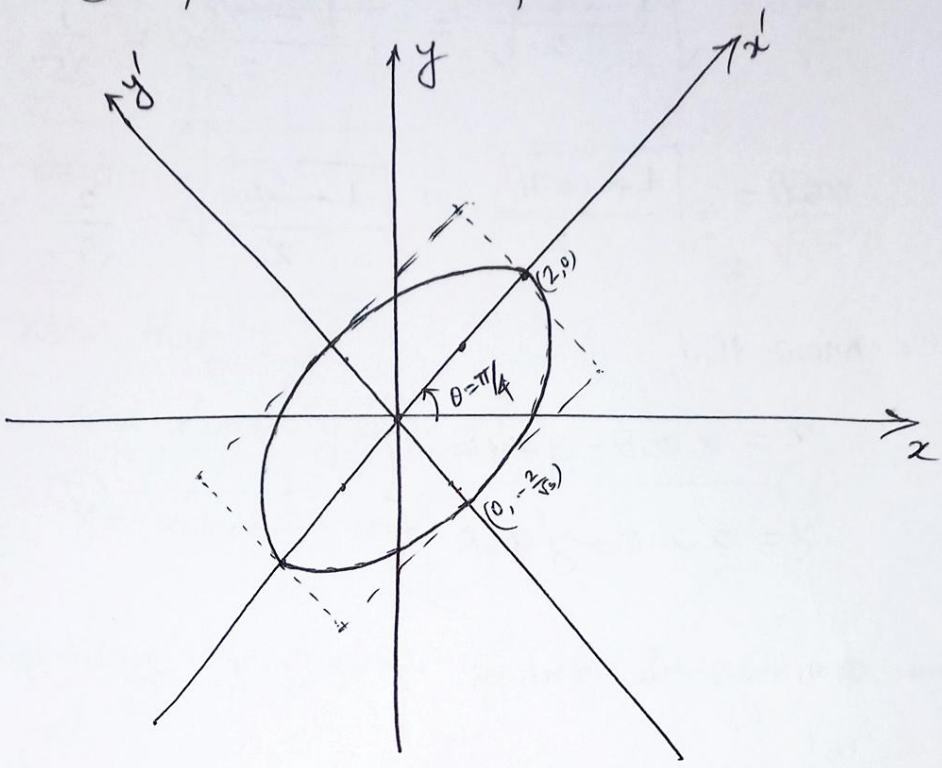
$$\Rightarrow \cancel{\frac{1}{2}x'^2} - \cancel{x'y'} + \frac{1}{2}y'^2 - \cancel{\frac{1}{2}x'^2} + \frac{1}{2}y'^2 + \frac{1}{2}x'^2 + \cancel{x'y'} + \frac{1}{2}y'^2 - 2 = 0$$

$$\Rightarrow \frac{1}{2}x'^2 + \frac{3}{2}y'^2 = 2$$

$$\Rightarrow \frac{x'^2}{4} + \frac{y'^2}{4/3} = 1$$

$$\Rightarrow \frac{x'^2}{(2)^2} + \frac{y'^2}{\left(\frac{2}{\sqrt{3}}\right)^2} = 1 \quad \text{——— (3)}$$

Equation (3) represents an ellipse.



Solution (iii):

Given that,

$$x^2+4xy-2y^2-6=0 \quad \text{---------- (1)}$$

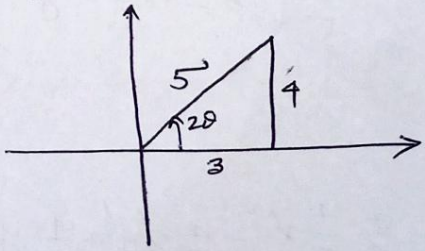
Comparing with the standard eqn we have

$$A=1,\ B=4,\ C=-2,\ F=-6$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{1+2}{4} = 3/4$$

$$\Rightarrow \cot 2\theta = 3/4$$

$$\therefore \cos 2\theta = 3/5$$



$$\therefore \sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1-3/5}{2}} = \frac{1}{\sqrt{5}}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1+3/5}{2}} = \frac{2}{\sqrt{5}}$$

We know that,

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (2)}$$

Substituting the values of $\sin\theta$ & $\cos\theta$ into equⁿ (2), we get

$$x = \frac{1}{\sqrt{5}}(2x' - y')$$

$$y = \frac{1}{\sqrt{5}}(x' + 2y')$$

Substituting the values of x & y into equⁿ (1) we have

$$\left\{\frac{1}{\sqrt{5}}(2x'-y')\right\}^2 + 4\cdot\frac{1}{5}(2x'-y')(x'+2y') - 2\left\{\frac{1}{\sqrt{5}}(x'+2y')\right\}^2 - 6 = 0$$

$$\Rightarrow \frac{1}{5}(4x'^2 - 4x'y' + y'^2) + \frac{4}{5}(2x'^2 - x'y' + 4x'y' - 2y'^2) - \frac{2}{5}(x'^2 + 4x'y' + 4y'^2) - 6 = 0$$

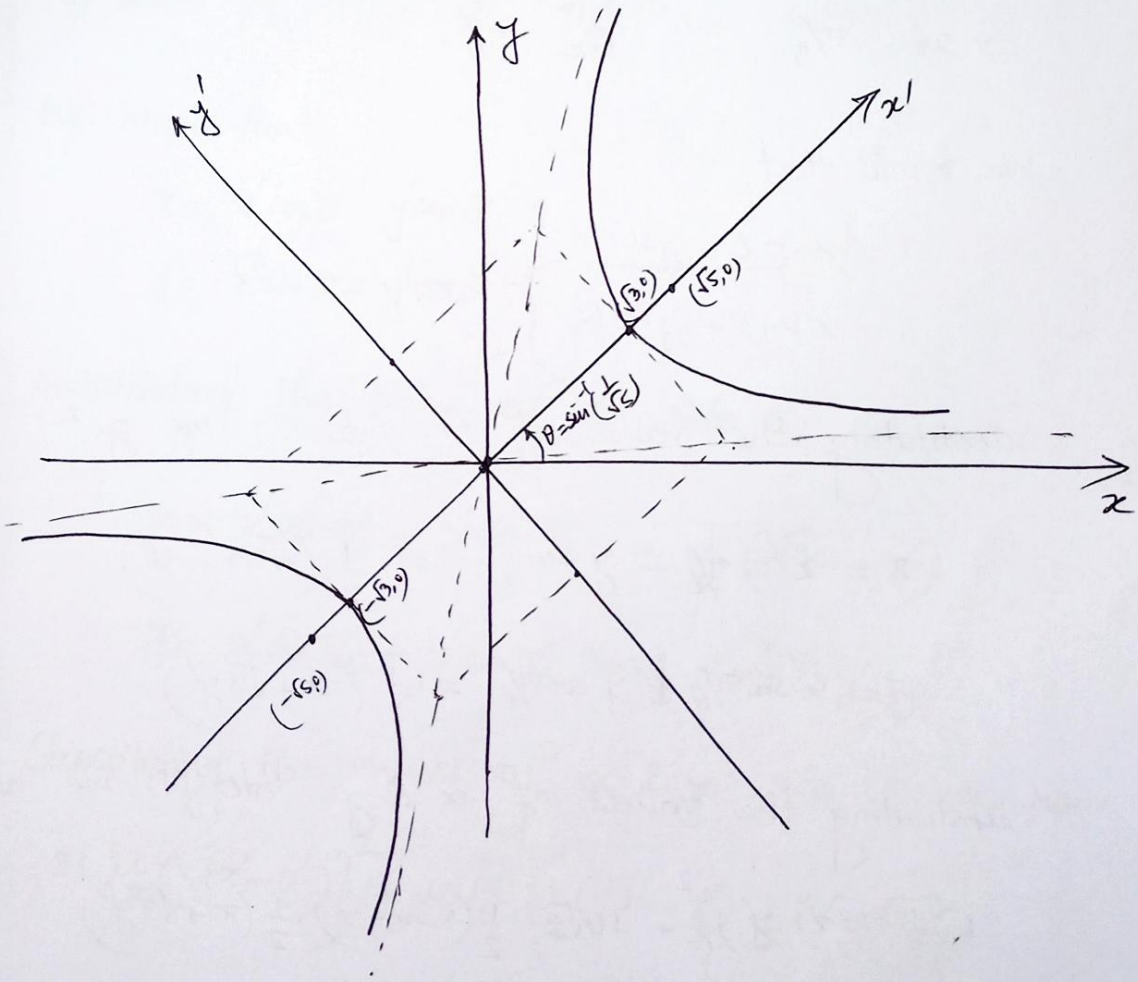
$$\Rightarrow \frac{4}{5}x'^2 - \frac{4}{5}x'y' + \frac{1}{5}y'^2 + \frac{8}{5}x'^2 + \frac{12}{5}x'y' - \frac{8}{5}y'^2 - \frac{2}{5}x'^2 - \frac{8}{5}x'y' - \frac{8}{5}y'^2 - 6 = 0$$

$$\Rightarrow 2x'^2 - 3y'^2 = 6$$

$$\Rightarrow \frac{x'^2}{3} - \frac{y'^2}{2} = 1$$

$$\Rightarrow \frac{x'^2}{(\sqrt{3})^2} - \frac{y'^2}{(\sqrt{2})^2} = 1 \quad \text{——— (3)}$$

Eqn (3) represents a hyperbola



Solution (iv) :

Given that,

$$31x^2 + 10\sqrt{3}xy + 21y^2 - 144 = 0 \quad \text{——— (1)}$$

Comparing with the standard eqn we get

$$A = 31,\ B = 10\sqrt{3},\ C = 21,\ F = -144$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{31-21}{10\sqrt{3}} = \frac{1}{\sqrt{3}} = \cot \pi/3$$

$$\Rightarrow 2\theta = \pi/3 \quad \therefore \quad \theta = \pi/6$$

We know that,

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (2)}$$

Substituting the value of θ into (2) we get

$$x = x'\cos\pi/6 - y'\sin\pi/6 = \frac{1}{2}(\sqrt{3}x' - y')$$

$$y = x'\sin\pi/6 + y'\cos\pi/6 = \frac{1}{2}(x' + \sqrt{3}y')$$

Substituting the values of x & y into (1) we have

$$31\left\{\frac{1}{2}(\sqrt{3}x' - y')\right\}^2 + 10\sqrt{3}\cdot\frac{1}{2}(\sqrt{3}x' - y')\cdot\frac{1}{2}(x' + \sqrt{3}y')$$

$$+ 21\left\{\frac{1}{2}(x' + \sqrt{3}y')\right\}^2 - 144 = 0$$

$$\Rightarrow \frac{31}{4}\left(3x'^2 - 2\sqrt{3}x'y' + y'^2\right) + \frac{5\sqrt{3}}{2}\left(\sqrt{3}x'^2 - x'y' + 3x'y' - \sqrt{3}y'^2\right)$$

$$+ \frac{21}{4}\left(x'^2 + 2\sqrt{3}x'y' + 3y'^2\right) - 144 = 0$$

$$\Rightarrow 31\left(3x'^2 - 2\sqrt{3}x'y' + y'^2\right) + 10\sqrt{3}\left(\sqrt{3}x'^2 + 2x'y' - \sqrt{3}y'^2\right)$$

$$+ 21\left(x'^2 + 2\sqrt{3}x'y' + 3y'^2\right) = 576$$

$$\Rightarrow 93x'^2 - \cancel{62\sqrt{3}x'y'} + 31y'^2 + 30x'^2 + \cancel{20\sqrt{3}x'y'} - 30y'^2$$

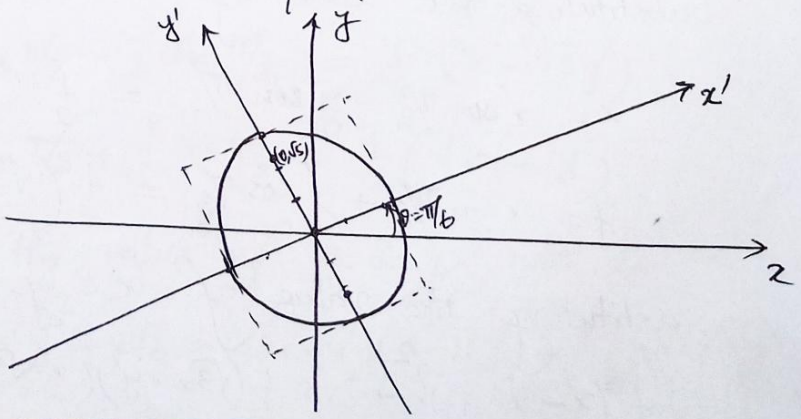
$$+ 21x'^2 + \cancel{42\sqrt{3}x'y'} + 63y'^2 = 576$$

$$\Rightarrow 144x'^2 + 64y'^2 = 576$$

$$\Rightarrow \frac{x'^2}{4} + \frac{y'^2}{9} = 1$$

$$\Rightarrow \frac{x'^2}{(2)^2} + \frac{y'^2}{(3)^2} = 1 \qquad \text{---------- (3)}$$

Eqn (3) represents an ellipse.



Solution (v):

Given that,

$$x^2+2\sqrt{3}xy+3y^2+2\sqrt{3}x-2y=0 \quad \text{——} (1)$$

Comparing with the standard eqn we get

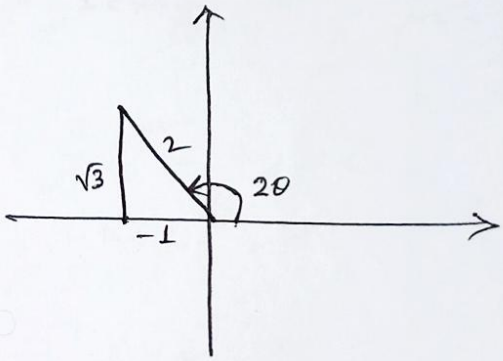
$$A=1,\ B=2\sqrt{3},\ C=3$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{1-3}{2\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

$$\therefore \cos 2\theta = -\frac{1}{2} = \cos\frac{2\pi}{3}$$

$$\Rightarrow 2\theta = \frac{2\pi}{3}$$

$$\Rightarrow \theta = \pi/3$$



We know that

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——} (2)$$

Substituting the value of θ into (2) we obtain

$$x = x'\cos\pi/3 - y'\sin\pi/3 = \frac{1}{2}(x'-\sqrt{3}y')$$

$$y = x'\sin\pi/3 + y'\cos\pi/3 = \frac{1}{2}(\sqrt{3}x'+y')$$

Substituting the value of x & y into (1) we have

$$\left\{\frac{1}{2}(x'-\sqrt{3}y')\right\}^2 + 2\sqrt{3}\cdot\frac{1}{4}(\sqrt{3}x'+y')(x'-\sqrt{3}y') + 3\left\{\frac{1}{2}(\sqrt{3}x'+y')\right\}^2$$

$$+2\sqrt{3}\cdot\frac{1}{2}(x'-\sqrt{3}y')-2\cdot\frac{1}{2}(\sqrt{3}x'+y')=0$$

$$\Rightarrow\frac{1}{4}(x'^2-2\sqrt{3}x'y'+3y'^2)+\frac{\sqrt{3}}{2}(\sqrt{3}x'^2+x'y'-3x'y'-\sqrt{3}y'^2)$$

$$+\frac{3}{4}(3x'^2+2\sqrt{3}x'y'+y'^2)$$

$$+\sqrt{3}x'-3y'-\sqrt{3}x'-y'=0$$

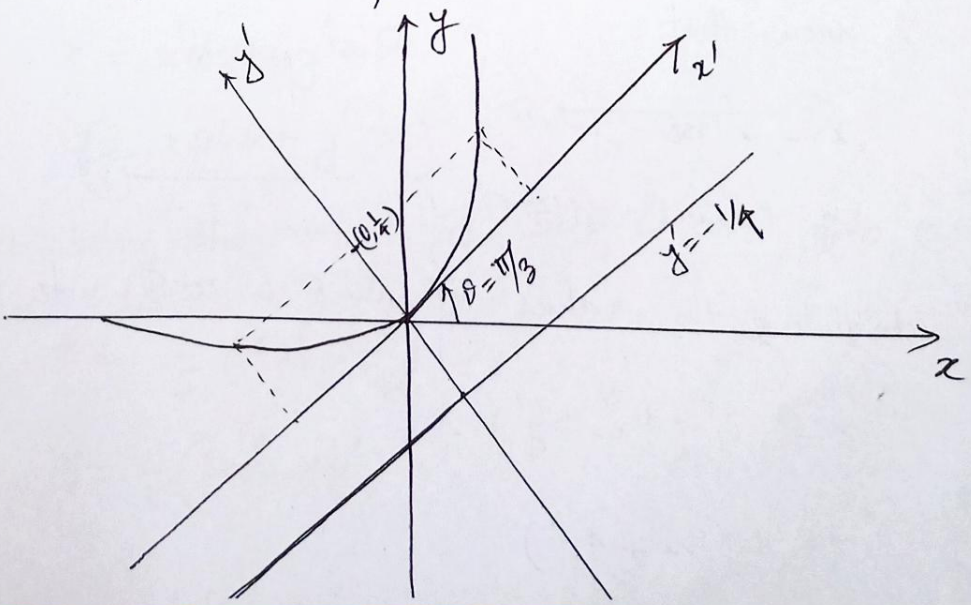
$$\Rightarrow\frac{1}{4}x'^2-\frac{\sqrt{3}}{2}x'y'+\frac{3}{4}y'^2+\frac{3}{2}x'^2-\sqrt{3}x'y'-\frac{3}{2}y'^2$$

$$+\frac{9}{4}x'^2+\frac{3\sqrt{3}}{2}x'y'+\frac{3}{4}y'^2-4y'=0$$

$$\Rightarrow 4x'^2-4y'=0$$

$$\Rightarrow x'^2=y'=4\cdot\left(\frac{1}{4}\right)y' \qquad \text{---------- (3)}$$

Equ^n (3) represents a parabola



Solution (vi):

Given that,

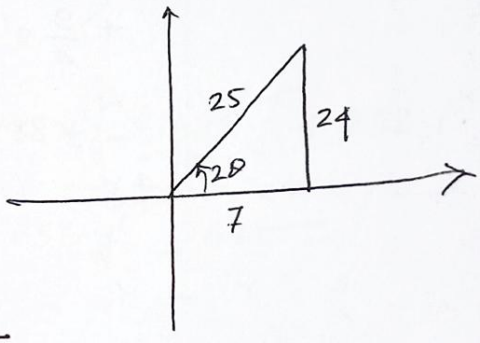
$$34x^2 - 24xy + 41y^2 - 25 = 0 \quad \text{——— (1)}$$

Comparing with the standard eqn we get.

$$A = 34,\ B = -24,\ C = 41,\ F = -25$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{34-41}{-24} = \frac{-7}{-24} = \frac{7}{24}$$

$$\therefore \cos 2\theta = 7/25$$



$$\therefore \sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = 3/5$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = 4/5$$

We know that

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (2)}$$

Substituting the values of $\sin\theta$ & $\cos\theta$ into (2) we obtain

$$x = \frac{1}{5}(4x' - 3y')$$

$$y = \frac{1}{5}(3x' + 4y')$$

Substituting the values of x & y into ① we obtain

$$34\left\{\frac{1}{5}(4x'-3y')\right\}^2 - \frac{24}{25}(4x'-3y')(3x'+4y') + 41\left\{\frac{1}{5}(3x'+4y')\right\}^2 - 25 = 0$$

$$\Rightarrow \frac{34}{25}(16x'^2 - 24x'y' + 9y'^2) - \frac{24}{25}(12x'^2 - 9x'y' + 16x'y' - 12y'^2) + \frac{41}{25}(9x'^2 + 24x'y' + 16y'^2) - 25 = 0$$

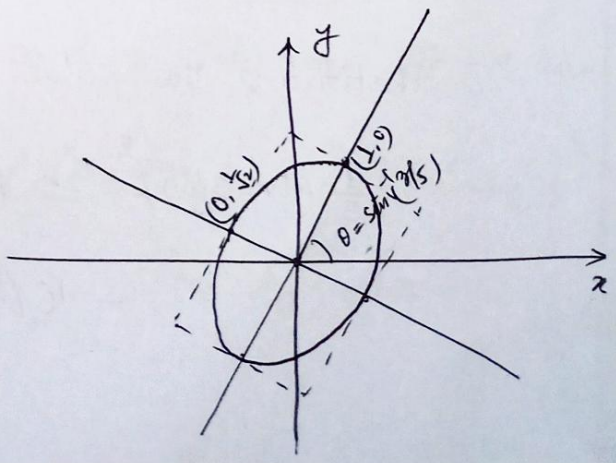
$$\Rightarrow 544x'^2 - 816x'y' + 306y'^2 - 288x'^2 - 168x'y' + 288y'^2 + 369x'^2 + 984x'y' + 656y'^2 = 625$$

$$\Rightarrow 625x'^2 + 1250y'^2 = 625$$

$$\Rightarrow x'^2 + 2y'^2 = 1$$

$$\Rightarrow \frac{x'^2}{(1)^2} + \frac{y'^2}{\left(\frac{1}{\sqrt{2}}\right)^2} = 1 \quad \text{——— ③}$$

Eqn ③ represents an ellipse



Solution (vii):

Given that,

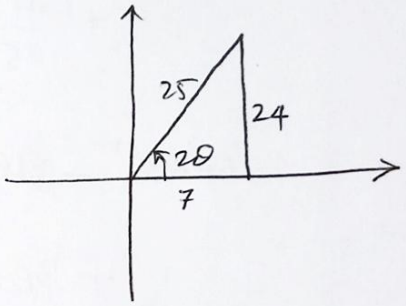
$$9x^2 - 24xy + 16y^2 - 80x - 60y + 100 = 0 \quad \text{---------- (I)}$$

Comparing with the standard equⁿ we get

$$A = 9,\ B = -24,\ C = 16,\ D = -80,\ E = -60,\ F = 100$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{9-16}{-24} = \frac{-7}{-24} = \frac{7}{24}$$

$$\therefore \cos 2\theta = \frac{7}{25}$$



$$\therefore \sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1-\frac{7}{25}}{2}} = \frac{3}{5}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1+\frac{7}{25}}{2}} = \frac{4}{5}$$

We know that

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta = \frac{1}{5}(4x' - 3y') \\ y &= x'\sin\theta + y'\cos\theta = \frac{1}{5}(3x' + 4y') \end{aligned}\right\} \quad \text{---------- (II)}$$

Substituting the values of x & y into equⁿ (I) we have

$$9\left\{\frac{1}{5}(4x'-3y')\right\}^2 - \frac{24}{25}(4x'-3y')(3x'+4y') + 16\left\{\frac{1}{5}(3x'+4y')\right\}^2 - 16(4x'-3y') - 12(3x'+4y') + 100 = 0$$

$$\Rightarrow \frac{9}{25}(16x'^2 - 24x'y' + 9y'^2) - \frac{24}{25}(12x'^2 - 9x'y' + 16x'y' - 12y'^2)$$

$$+\frac{16}{25}(9x'^2 + 24x'y' + 16y'^2) - 64x' + 48y' - 36x' - 48y' + 100 = 0$$

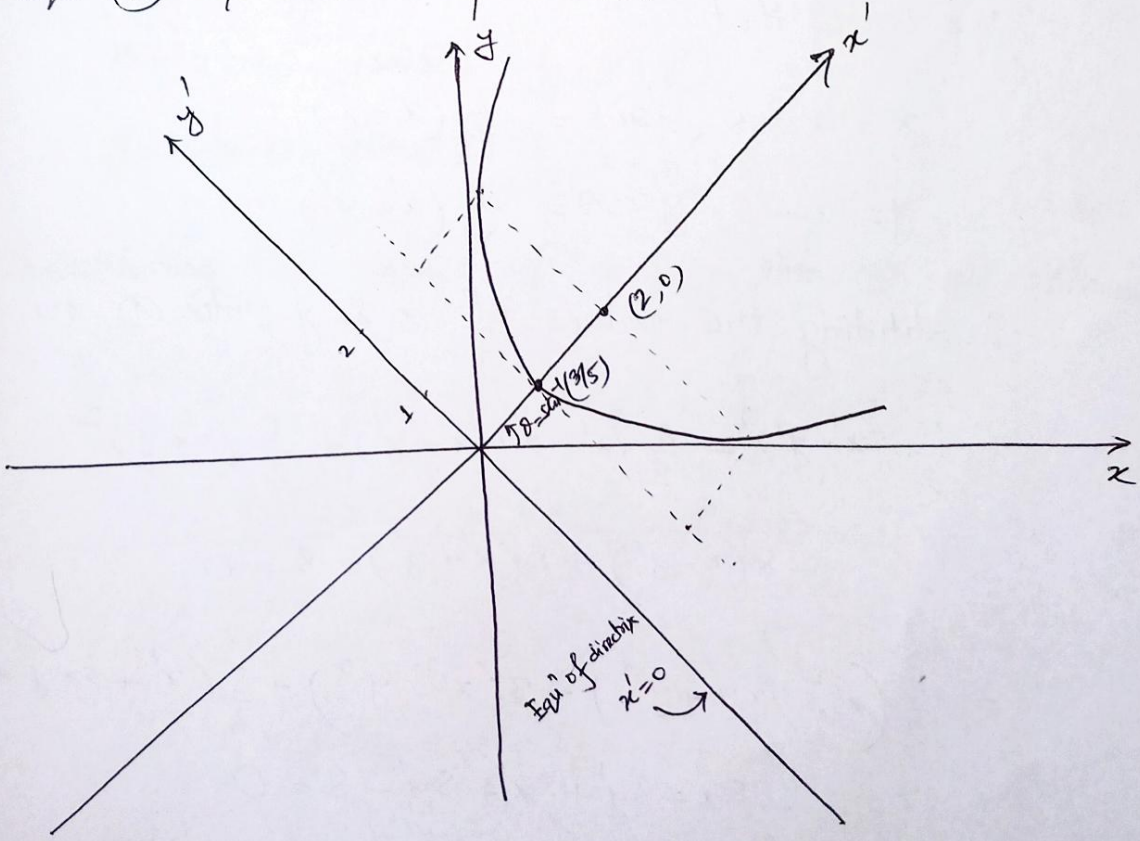
$$\Rightarrow 144x'^2 - 216x'y' + 81y'^2 - 288x'^2 - 168x'y' + 288y'^2$$

$$+144x'^2 + 384x'y' + 256y'^2 - 2500x' + 2500 = 0$$

$$\Rightarrow 625y'^2 = 2500(x' - 1)$$

$$\Rightarrow y'^2 = 4.1(x' - 1) \quad \text{——— (III)}$$

Eqn (III) represents a parabola.



Solution (viii):

Given that,

$$5x^2 - 6xy + 5y^2 - 8\sqrt{2}x + 8\sqrt{2}y = 8 \quad \text{——— (1)}$$

Comparing with the standard eqn we get

$$A = 5,\ B = -6,\ C = 5,\ D = -8\sqrt{2},\ E = 8\sqrt{2},\ F = -8$$

$$\therefore \cot 2\theta = \frac{A - C}{B} = \frac{5-5}{-6} = 0 = \cot \pi/2$$

$$\Rightarrow 2\theta = \pi/2$$

$$\Rightarrow \theta = \pi/4$$

We know that,

$$x = x'\cos\theta - y'\sin\theta = \frac{1}{\sqrt{2}}(x' - y')$$

$$y = x'\sin\theta + y'\cos\theta = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x & y into (1) we obtain

$$\frac{5}{2}(x'-y')^2 - \frac{6}{2}(x'-y')(x'+y') + \frac{5}{2}(x'+y')^2$$

$$-8(x'-y') + 8(x'+y') - 8 = 0$$

$$\Rightarrow \frac{5}{2}(x'^2 - 2x'y' + y'^2) - 3(x'^2 - y'^2) + \frac{5}{2}(x'^2 + 2x'y' + y'^2)$$

$$-8x' + 8y' + 8x' + 8y' - 8 = 0$$

$$\Rightarrow 5x'^2 - 10x'y' + 5y'^2 - 8x'^2 + 6y'^2 + 5x'^2 + 10x'y' + 5y'^2 + 32y' - 16 = 0$$

$$\Rightarrow 4x'^2 + 16y'^2 + 32y' = 16$$

$$\Rightarrow x'^2 + 4y'^2 + 8y' = 4$$

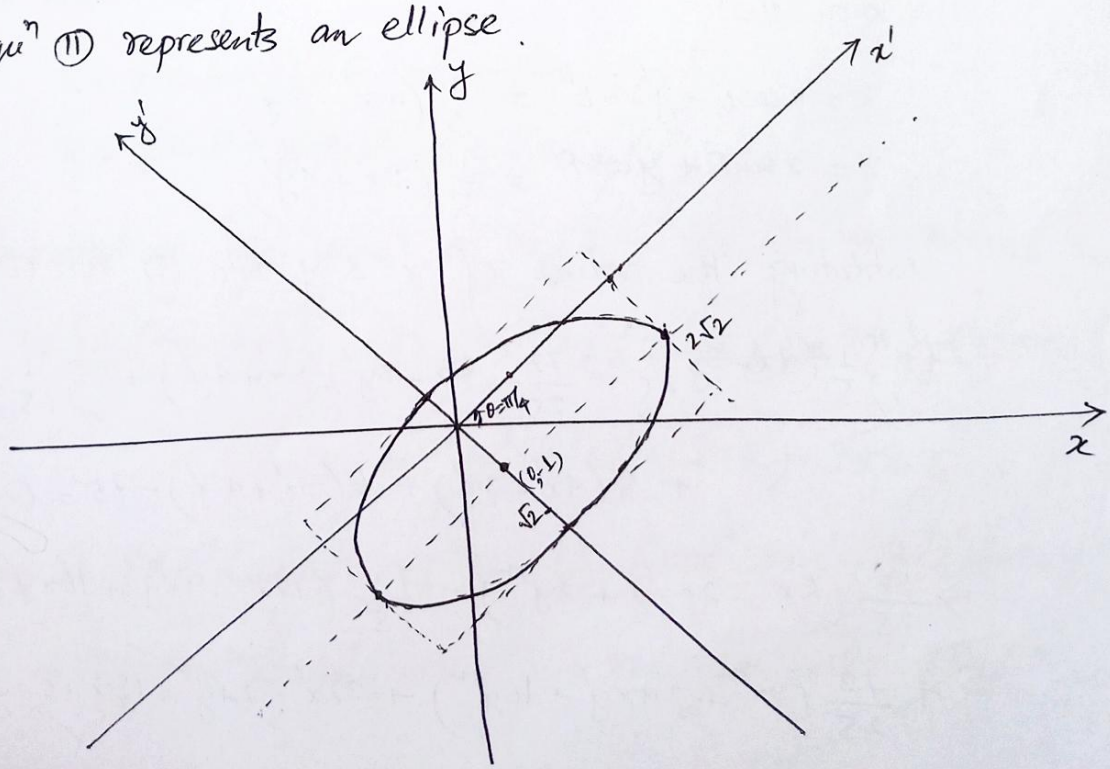
$$\Rightarrow x'^2 + 4(y'^2 + 2y' + 1) - 4 = 4$$

$$\Rightarrow x'^2 + 4(y'+1)^2 = 8$$

$$\Rightarrow \frac{x'^2}{8} + \frac{(y'+1)^2}{2} = 1$$

$$\Rightarrow \frac{x'^2}{(2\sqrt{2})^2} + \frac{(y'+1)^2}{(\sqrt{2})^2} = 1 \quad \text{——— (11)}$$

Eqn (11) represents an ellipse.



Solution (ix):

Given that,

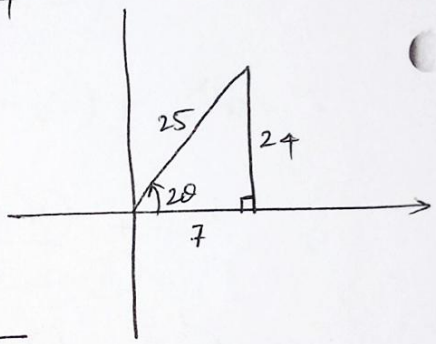
$$52x^2 - 72xy + 73y^2 + 40x + 30y - 75 = 0 \quad \text{---(1)}$$

Comparing with the standard eqn we get

$$A = 52,\ B = -72,\ C = 73,\ D = 40,\ E = 30,\ F = -75$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{52-73}{-72} = \frac{7}{24}$$

$$\therefore \cos 2\theta = \frac{7}{25}$$



$$\therefore \sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = \frac{3}{5}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1+7/25}{2}} = \frac{4}{5}$$

We know that,

$$x = x'\cos\theta - y'\sin\theta = \frac{1}{5}(4x' - 3y')$$

$$y = x'\sin\theta + y'\cos\theta = \frac{1}{5}(3x' + 4y')$$

Substituting the values of x & y into (1) we obtain

$$52\left\{\frac{1}{5}(4x'-3y')\right\}^2 - \frac{72}{25}(4x'-3y')(3x'+4y') + 73\left\{\frac{1}{5}(3x'+4y')\right\}^2 + 8(4x'-3y') + 6(3x'+4y') - 75 = 0$$

$$\Rightarrow \frac{52}{25}(16x'^2 - 24x'y' + 9y'^2) - \frac{72}{25}(12x'^2 - 9x'y' + 16x'y' + 12y'^2) + \frac{73}{25}(9x'^2 + 24x'y' + 16y'^2) + 32x' - 24y' + 18x' + 24y' - 75 = 0$$

$$\Rightarrow 832x'^2 - 1248x'y' + 468y'^2 - 864x'^2 - 504x'y' + 864y'^2 + 657x'^2 + 1752x'y' + 1168y'^2 + 1250x' - 1875 = 0$$

$$\Rightarrow 625x'^2 + 1250x' + 2500y'^2 = 1875$$

$$\Rightarrow x'^2 + 2x' + 4y'^2 = 3$$

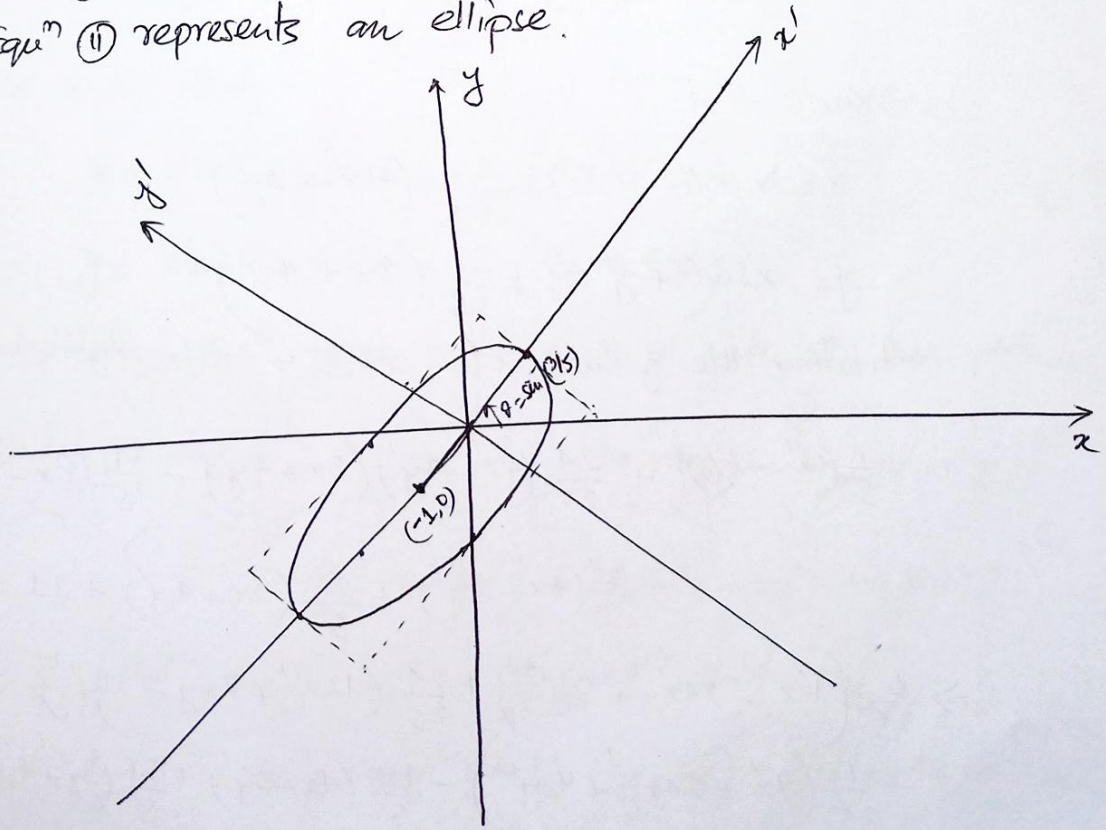
$$\Rightarrow x'^2 + 2x' + 1 + 4y'^2 = 3 + 1 = 4$$

$$\Rightarrow (x'+1)^2 + 4y'^2 = 4$$

$$\Rightarrow \frac{(x'+1)^2}{4} + y'^2 = 1$$

$$\Rightarrow \frac{(x'+1)^2}{(2)^2} + \frac{y'^2}{(1)^2} = 1 \quad \text{---------- (11)}$$

Eqn (11) represents an ellipse.



Solution (x):

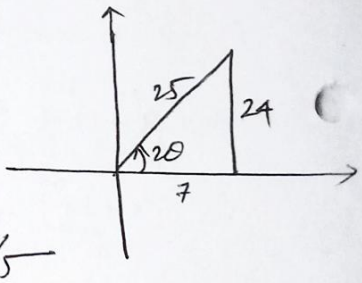
Given that,

$$6x^2+24xy-y^2-12x+26y+11=0 \quad \text{——— (1)}$$

Comparing with the standard eqn we get

$$A=6,\ B=24,\ C=-1,\ D=-12,\ E=26,\ F=11$$

$$\therefore \cot 2\theta = \frac{A-C}{B} = \frac{6+1}{24} = 7/24$$



$$\therefore \cos 2\theta = 7/25$$

$$\therefore \sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = 3/5$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1+7/25}{2}} = 4/5$$

We know that,

$$x = x'\cos\theta - y'\sin\theta = \frac{1}{5}(4x'-3y')$$

$$y = x'\sin\theta + y'\cos\theta = \frac{1}{5}(3x'+4y')$$

Substituting the values of x & y into (1) we obtain

$$6\left\{\frac{1}{5}(4x'-3y')\right\}^2 + \frac{24}{25}(4x'-3y')(3x'+4y') - \left\{\frac{1}{5}(3x'+4y')\right\}^2$$

$$-\frac{12}{5}(4x'-3y') + \frac{26}{5}(3x'+4y') + 11 = 0$$

$$\Rightarrow \frac{6}{25}(16x'^2 - 24x'y' + 9y'^2) + \frac{24}{25}(12x'^2 + 7x'y' - 12y'^2)$$

$$-\frac{1}{25}(9x'^2 + 24x'y' + 16y'^2) - \frac{12}{5}(4x'-3y') + \frac{26}{5}(3x'+4y') + 11 = 0$$

$$\Rightarrow 96x'^2 - 144x'y' + 54y'^2 + 288x'^2 + 168x'y' - 288y'^2$$
$$- 240x' + 180y' + 390x' + 520y' + 275 = 0$$
$$- 9x'^2 - 24x'y' - 16y'^2$$

$$\Rightarrow 375x'^2 + 150x' - 250y'^2 + 700y' + 275 = 0$$

$$\Rightarrow 15x'^2 + 6x' - 10y'^2 + 28y' + 11 = 0$$

$$\Rightarrow 15\left(x'^2 + \frac{2}{5}x'\right) - 10\left(y'^2 - \frac{14}{5}y'\right) + 11 = 0$$

$$\Rightarrow 15\left\{x'^2 + 2 \cdot x' \cdot \frac{1}{5} + \left(\frac{1}{5}\right)^2 - \left(\frac{1}{5}\right)^2\right\}$$
$$- 10\left\{y'^2 - 2 \cdot y' \cdot \frac{7}{5} + \left(\frac{7}{5}\right)^2 - \left(\frac{7}{5}\right)^2\right\} + 11 = 0$$

$$\Rightarrow 15\left\{\left(x' + \frac{1}{5}\right)^2 - \frac{1}{25}\right\} - 10\left\{\left(y' - \frac{7}{5}\right)^2 - \frac{49}{25}\right\} + 11 = 0$$

$$\Rightarrow 15\left(x' + \frac{1}{5}\right)^2 - \frac{3}{5} - 10\left(y' - \frac{7}{5}\right)^2 + \frac{98}{5} + 11 = 0$$

$$\Rightarrow 15\left(x' + \frac{1}{5}\right)^2 - 10\left(y' - \frac{7}{5}\right)^2 + 30 = 0$$

$$\Rightarrow \frac{\left(x' + \frac{1}{5}\right)^2}{2} - \frac{\left(y' - \frac{7}{5}\right)^2}{3} + 1 = 0$$

$$\Rightarrow \frac{(y' - 7/5)^2}{(\sqrt{3})^2} - \frac{(x' + 1/5)^2}{(\sqrt{2})^2} = 1 \quad \text{——— (11)}$$

Eqn (11) represents a hyperbola.

